Jayanth Dasamantharao

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway Associates, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Built and deployed production GenAI systems using LLMs (GPT-4, LLaMA, Gemini), RAG pipelines, and semantic embeddings, improving response accuracy by 30%+ and reducing latency by 25%.
- Designed ML infrastructure with FastAPI/Flask, cloud endpoints, Git-based CI workflows, and POCs transitioned into live customer portals. Evaluated/monitored using metrics like F1, AUC-ROC, A/B testing.
- Implemented human-in-the-loop, multimodal AI systems, enabling text + image retrieval/generation, explainability, and escalation for non-technical users. Optimized inference using embedding reuse, vector databases.
- Built scalable recommender, analytics, and chatbot systems using Python, Duckdb, SQL and React/Dash/Plotly, processing 5M+ rows to drive customer profiling, churn prediction, and 30% higher engagement.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python data processing and evaluation pipelines, implementing hash maps, sliding windows, priority queues, batching strategies to aggregate, validate, analyze data, improving 40% engagement quality.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, and indexed lookups to efficiently clean, deduplicate, and transform large datasets, reducing production issues by 50%.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming: Python, R, C++, SQL. **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
ML / AI : Model Deployment, Evaluation, Monitoring, Debugging, Feature Engineering.
GenAI & LLMs: LLM fine-tuning, RAG, Prompt Engineering, LangChain, Vector Databases, Multimodal AI.
Cloud & MLOps: PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Pinecone, Chroma.
Deep Learning: Neural Networks, CNNs, LSTMs, YOLO, Faster R-CNN, GANs, Time-Series Forecasting.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ
Personal Website: https://jayanthdasamantharao.github.io/

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

Multimodal GenAI Chatbot (Production): LLM-powered chatbot with RAG, embeddings, caching, and text + image outputs, deployed on cloud APIs along with Human-In-The-Loop architecture and optimization.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: https://github.com/JayanthDasamantharao